

ARNAV VERMA

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EDUCATION

New York University, Courant Institute of Mathematical Sciences – New York, NY

Expected: 05/28

*Master of Science in **Computer Science***

Affiliations: Tech@NYU, HackNYU

Coursework: Fundamental Algorithms, Machine Learning, Deep Learning

University of North Texas — Denton, TX

01/23 – 05/26

*Bachelor of Science in **Computer Science, Mathematics Minor***

GPA: 3.70

Affiliations: IEEE Computer Society, UNT AI Research Program

Awards: President's List, Dean's List, **Undergraduate Research Fellowship, International Education Scholarship**

Coursework: Algorithms (A+), Artificial Intelligence (A+), Natural Language Processing (A+), Linear Algebra (A+)

PUBLICATIONS

Iqbal, F., Dutt, R., Das, S., **Verma, A.**, Ray Choudhury, S. "Evaluating Multi-Dimensional Generalization of Large Language Models in Temporal Extraction Tasks." **Under review, ACL ARR 2026.**

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Typescript

Libraries: PyTorch, TensorFlow, JAX, HuggingFace, Scikit-learn, Flask

Tools: Docker, Kubernetes, Terraform, PostgreSQL, MongoDB, Google AI Studio

Concepts: AI/ML, Deep Learning, Software Development, DevOps, API Development

EXPERIENCE

Undergraduate Research Fellow

05/25 - Present

University of North Texas - Denton, TX

- Conducted a systematic multi-dimensional generalization study of LLMs (LLaMA, Qwen, Mistral: Dense and MoE) across **42 model configurations** (3 families × 2 architectures × 7 reasoning strategies) on time and event expression extraction across 4 generalization dimensions: domain shift, adversarial robustness, compositionality, and structural length
- Built evaluation pipelines using Hungarian word level LCS matching for F1 scoring, adversarial datasets via SeqAttack (**TextFooler/DeepWordBug**), and a clinical domain-shift corpus (i2b2 T2C) revealing negative Spearman rank correlation (**$\rho = -0.15$**) between clinical and all other dimensions for **events extractions**
- Fine-tuned RoBERTa, T5, and GPT-2 baselines for **time extractions** and **event extractions** sequence labeling (BIO tagging, HuggingFace Trainer); showed fine-tuned PLMs hold more stable cross-dimensional rankings than prompted LLMs despite lower peak performance

PROJECTS

Polish – Typescript, Express, React, PostgreSQL, Redis, Cloudflare R2, Google AI Studio

[Demo](#) | [Code](#)

- Engineered a full-stack AI-powered resume SaaS using Next.js 14, Express.js, PostgreSQL, and Redis, integrating both Claude (Anthropic) and **Gemini 2.5 Flash** APIs with user-provided API key support, supporting grammar, ATS optimization, clarity, and content suggestions across **5+ improvement categories**
- Built a complete version control system with full content snapshots, paginated history, line by line diff comparison, and one click restoration, backed by an indexed PostgreSQL schema and a debounced autosave that flushes to localStorage on failure to prevent data loss
- Shipped production infrastructure on Railway with Dockerized multi-stage builds, GitHub Actions CI (type-check + build gates), dual JWT authentication (**15-min access / 7-day refresh with bcrypt-12 hashing**), and multi-format document export (PDF, DOCX, LaTeX) using client-side parsing libraries

Marigold – Python, FastAPI, React, Typescript, Cloudflare, MongoDB, **Google AI Studio**

[Demo](#) | [Code](#)

- Built a full-stack AI study app using React, FastAPI, and MongoDB, featuring PDF upload, **Google Gemini 2.5 Flash** powered flashcard generation, and a timed quiz mode with scoring and answer review
- Engineered a secure JWT authentication system with access/refresh token rotation, httpOnly cookies, and bcrypt password hashing, protecting **8+ REST API** endpoints across **4 resource domains**
- Designed an end-to-end study workflow from PDF parsing (PyMuPDF) to AI-generated multiple-choice flashcards with distractors, reducing manual note-review time by enabling on-demand quiz sessions from any uploaded document

AWARDS & HACKATHONS

1st Place: HackSMU '24 & '25, HackUNT '23 & '24 **2nd Place:** HackUTD '24 **Runner-up:** HackTX (UT Austin)